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| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 01/18/2019 |
| **PI Name:** | Pierce, Mark Ph.D. | | |
| **Project Title:** | Short Wave Infrared (SWIR) Imaging of Lymphatics | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | TRISH--TRISH | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | mp941@soe.rutgers.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | |
| **Organization Name:** | Rutgers, The State University of New Jersey | | |
| **PI Address 1:** | Department of Biomedical Engineering | | |
| **PI Address 2:** | 599 Taylor Road | | |
| **PI Web Page:** | | | |
| **City:** | Piscataway | **State:** | NJ |
| **Zip Code:** | 08854 | **Congressional District:** | 6 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | TRISH--Synergy |
| **Start Date:** | 02/01/2018 | **End Date:** | 09/30/2018 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | TRISH |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Sevick, Eva Ph.D. ( University of Texas Health Science Center ) | | |
| **Grant/Contract No.:** | NNX16AO69A-SYN0001 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Synergy Project<br>This project builds upon an existing TRISH (Translational Research Institute for Space Health) application at the University of Texas Health Science Center Houston (UTHSCH) with Principal Investigator (PI) Eva Sevick, Ph.D. to extend the wavelengths of lymphatic imaging into the short wave infrared (SWIR) spectral region for greater penetration depth to visualize cerebrospinal fluid outflow into the lymphatics in a large animal model. Feasibility of SWIR imaging will be established with rare-Earth albumin encapsulated nanoparticles (REANPs) intrathecally injected in swine already investigated by TRISH-funded UTHSCH, with near-infrared fluorescence (NIRF) imaging for direct comparison. | | |
| **Rationale for HRP Directed Research:** | | | |

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| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | Rare-Earth-doped nanocomposites were synthesized, characterized, and encapsulated in albumin at Rutgers University, prior to shipment to UTHSC. Preliminary cell viability and animal imaging were performed at Rutgers in conjunction with ongoing studies on rare-Earth albumin encapsulated nanoparticles (REANPs). Initial findings confirmed that REANCs could be synthesized with a narrow range of physical sizes and rendered biocompatible for pilot studies in cells and animals. |
| **Bibliography Type:** | Description: (Last Updated: ) |